An Autoencoder Intrusion Detection System for sEMG: A Detection-Based Alternative for EMG Security

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***Abstract*—As surface Electromyography (sEMG) technology grows in popularity and use cases, from medical diagnostics to consumer wearables and IoT interfaces, the attack surface for malicious injection and impersonation grows. Current research primarily focuses on preventive authentication, leaving a critical gap for post-prevention Intrusion Detection Systems (IDS) capable of identifying anomalies in real-time. This paper proposes a lightweight Convolutional Autoencoder-based IDS trained on Root Mean Square (RMS) heatmaps to detect unauthorized users through the difference between an authorized user's trained sEMG signature and an intruder's unique, but reconstruction-error-prone, sEMG signature. Using the Ninapro DB5 dataset, we developed a model under 11 KB in size that operates on a minimal sample window of 40 data points, significantly reducing latency compared to prior RMS-based methods. Our results demonstrate the feasibility of this approach, achieving an intruder detection rate of up to 61.1% and a false rejection rate of 20.4% on holdout data. These findings suggest that ultra-lightweight convolutional autoencoder models can effectively serve as a secondary security layer for resource-constrained wearable sEMG devices.**

I. Introduction

Electromyography (EMG) is a medically recognized diagnostic test that evaluates the health of skeletal muscles and the nerves that control them. In the same medical context, this test is often performed by a neurologist or psychiatrist by inserting small electrodes into the muscles being tested, and then by recording and analyzing the electrical activity of these muscles. Over time, a newer and less invasive form of EMG was made, called surface electromyography (sEMG), where electrodes are placed on the skin to measure the electrical activity of muscles, generated by muscle fiber contractions.

As the concept of sEMG has evolved, so have the instruments needed to perform it. Modern innovations allow for a much smaller scale of devices, such as tech-embedded wearable sEMG wristbands, to use non-invasive, higher sensitivity electrodes to capture and process electrical signals generated by muscles and digitally transmit the data to another machine. The data used from these wristbands, thanks to their mobile nature, are now also used in much more than just medical diagnostics, in tasks like device interaction via biofeedback as control interfaces in augmented reality smart lens and virtual reality headsets, and much wider muscle activity research via non-invasive sEMG measurements.

However, as the field of sEMG research utilizes smarter, smaller-scale, and more efficient technologies, the need to be wary of malicious actions and exploitations of the unique security issues that sEMG wearables present still persists. Similarly, as other types of extremely personal health data, such as biomarkers, body/neural signals, genetic data, and other bioinformation should be held to the highest privacy standards, sEMG signals are no different. The safety and privacy of these signals remain an extremely high priority, especially as the use of sEMG technology grows in popularity and finds itself as another attack surface for malicious actors to interface with and connect with personal electronics of individuals like mobile phones and smartwatches.

This paper aims to propose a lightweight convolutional autoencoder trained and evaluated on RMS (Root Mean Square)-converted EMG maps as a means of intrusion detection for EMG collecting systems, primarily evaluated in this paper by its means of detecting impersonation through unique machine-identified EMG signatures in these maps.

Ali Elgazzar worked on proposing our initial concepts, finding relevant papers and research that supported our work, security ideas, and writing out the introduction and related works sections. Davis Zhong worked on finding different ways to credibly validate our results as well as creating visualizations, and finding alternate implementations that could improve upon

our initial results via better subject differentiation, as well as writing the results and conclusion sections. Aditya Pillai worked on creating an initial proof of concept and initial prototypes, creating visualizations for validation tests, and writing the abstract and methodology section.

II. Related Work

This research paper will discuss previous works related to wearable EMG sensors and its security in four sections: Structure of sEMG, sEMG applications, sEMG security issues and attacks, and proposed mitigations to sEMG issues.

A. Structure of sEMG

While the ability to read signals moving through the muscles of the human body and tying that to actions has been crucial in the medical space, translating that information into a consumer based input method requires a different structure. *Sul et al.* [1] provides a guideline for such a structure, consisting of acquisition of signals from sEMG devices, filtering of noise for a better defined signal, and analysis of sEMG signals to determine muscle movement and map to command. Despite the intended application of an exoskeleton application, this pipeline can be applied to most applications that use sEMG as an input mechanism.

B. Applications of sEMG

With this framework in mind for sEMG structures, many researchers have proposed uses for sEMG devices as a unique input method for consumer scenarios.

A recent study by *Muthulakshmi et al.* [2] shows one such use: Hand Gesture Recognition. In general, the authors state that the study aimed "to recognize and classify specific hand gestures related with the desired functions." This is done by reading sEMG signals from the wrist/forearm of a user using a wearable sEMG sensor. After each different gesture, such as a fist or a peace sign, is classified with the help of machine learning models such as Support Vector Machine (SVM) and K-Nearest Neighbors (KNN). With these classifications, they are mapped to different calls or functions of a mobile application, essentially being inputs. Given the further applications of an input method like this (such as accessibility for controlling IoT devices), ensuring that these inputs are integral and confidential is key for future structures using this technology.

Another approach of using sEMG is as a Thing-You-Are authentication factor. *Shin et al.* [3] outlines the general method of detecting sEMG signals unique to a user through machine learning and using that as a personal authentication key. This authentication method has the benefit of not needing to be memorized and can't be easily lost since they are tied to a user's person.

C. EMG security issues and attacks

sEMG has been shown to have applications in multiple different scenarios, allowing for streamlined control of apps and IoT devices as well as an accessibility option. For example, the Mudra Link wristband allows for human computer interaction similar to Meta's neural interface, but for most mobile devices instead of the Meta RayBan Displays. However, this technology has been proven to have vulnerabilities, as with most other technologies. The notable attacks that researchers have provided are Injections (or radio adversarial attacks) and key-logging side-channel attacks.

Similar to IP spoofing, sEMG signals can be fabricated and sent to a gateway, claiming to be part of the legitimate sEMG sensor. The attack, proposed by *H. Xie* [4], uses radio frequency to bypass traditional encryption and allow for a direct injection of signals. Other than the fact that these frequencies cause the jamming of legitimate signals, the attack demonstrated can also facilitate remote interference and control of sEMG structures, specifically a gesture based sEMG application, one of the more popular directions of sEMG in the consumer market. The effects of this attack can range from denial of service to injections and hijacks.

However, sEMG structures are also susceptible to passive attacks as well. Specifically, keylogging is possible through learning arm/hand movements and mapping them to keys and actions, helping a bad actor to learn passwords and other important information. This attack, demonstrated by *Gazzari et al.* [5], takes advantage of the fact that most wearable devices, including sEMG sensors, are made to be worn most of the day or all day. As the authors stated, "[Wearable devices] will inevitably be present when the wearers engage in sensitive activities, gathering data that may contain traces of these activities." With this in mind, an attacker can use the wearable sEMG device to act as a keylogger to read strokes of users when they interact with any device throughout their day, as long as they have the infected device on them. This experiment considered different ages, genders, and typing methods of 37 participants, further putting evidence that this attack is feasible to most users.

Given that these threats have been proven to be real, researchers have put efforts in preventing these attacks from impacting users.

D. Proposed Mitigations for sEMG issues

Previous work in the research of sEMG security mitigations is somewhat broad, but there are a number of works that touch on different implementations for more secure sEMG architectures.

Firstly, a two-factor authentication system is proposed by *Li et al.* [6]. They employed machine learning using One-Class Support Vector Machine (OCSVM) and Local Outlier Factor (LOF) classifiers to analyze personal patterns from the movement of a subject's forearm muscle. This approach successfully prevented unauthorized access by detecting differences in sEMG signals during pattern entry. This work is of particular interest as it demonstrates the potential of sEMG as a biometric authenticator and offers an effective solution against shoulder surfing and side-channel attacks.

On the network side, secure communication channels can be achieved through creating short-lived session keys from sEMG

signals. By utilizing two proximate sensors on the same muscle, *Yang et al.* [7] generated matching unique codes to enable short-range authentication. Similarly, *Usman et al.* [8] offers a convolutional neural network-based contaminant identification method to clean sEMG data without manual feature extraction which, in addition to outperforming traditional support vector machines, was also able to identify multiple contaminants like power line interference, motion artifacts, ECG cross-talk, amplitude saturation, and additive white Gaussian noise. These papers were notable in showing how network security can be achieved with lightweight CNN to filter and clean sEMG data and protection against eavesdropping via unauthorized pairing with sEMG-based cryptographic

Lu et al. [9] explored multiple approaches, including Discrete Wavelet Transform (DWT) with ExtraTreesClassifier, Continuous Wavelet Transform (CWT) with Convolutional Neural Networks (CNN), transfer learning, and Siamese networks, to identify and verify users from surface sEMG signals. The DWT with ExtraTreesClassifier method achieved a 99.21% accuracy from but is limited in that it requires full retraining for new users, while the CWT with CNN approached converted 1D sEMG signals into 2D time-frequency images, achieving, 99.2% accuracy with automated feature extraction. Transfer learning enabled scalable addition of new users with minimal retraining, maintaining over 90% accuracy, and Siamese networks allowed verification of previously unseen users, demonstrating adaptability in dynamic Internet of Things (IoT) scenarios. These results highlight the power of sEMG signals for both identification and verification in authentication roles, supporting its use as a method of continuous monitoring and screening for adversarial attacks.

Noise, in addition, is another security issue in sEMG systems. *McCool et al.* [10] developed a system to identify sources of such noise in sEMG signals. The system was able to classify noise between ECG interference, motion artifacts, power line interference, amplifier saturation, and additive white Gaussian noise. The one-class SVM approach was very effective at classifying between contaminant types at low Signal-to-Noise Ratios (SNR). This system's ability to differentiate between contaminant sources is of interest when employing targeted correction strategies, where knowledge of a specific type of contaminant can help a system to perform specific kinds of corrective measures. Additionally, the classification system introduced Signal-to-Powerline Ratio, Signal-to-ECG Ratio, and Correlation Coefficient for Normality as trainable metrics for their model as a means of detecting other anomalies , identifying them as false positives or intentional interference.

Lastly, *Li et al.* [11] proposed a deep anomaly detection framework that is capable of detecting imposter activity across multiple gestures and channels, outperforming single channel methods. Compared to other sEMG biometric frameworks, such as those using Discrete Wavelet Transform (DWT) with ExtraTreesClassifier or Siamese networks for multi-user identification, this method differs by focusing on per-user anomaly detection. Instead of feeding time-series sEMG data into their detection model, they transform the data into 2D sEMG maps, effectively turning their task of signal analysis into a visual analysis task. This paper discovered that images converted via Root Mean Square (RMS) transformations had better performance in capturing spatial distribution of muscle energy, and by converting data to an RMS map, a unique topographic map can be created where even if two people perform the same gesture. Furthermore, this RMS map technique achieved success in high-security authentication in systems with sparse sensors, allowing for the use of this anomaly detection framework with limited hardware.

The mitigation techniques provided by previous works does show that much can be done in securing the sEMG field. From secure communication to proper authentication, the ideas and implementations proposed have shown to address the previously mentioned issues and attacks that are particular in sEMG systems. However, the factor that these papers have in common is that they approach the security of sEMG in a preventive manner. While preventive security is effective in mitigating most attacks, they lack defense ability when an attack is successful in a system or if an attack is done in compliance, such as an attack done by an insider. To address this, detection methods for security are needed in conjunction with the discussed preventive measures.

III. Methodology

Our main goal is to provide an IDS system that can eventually be tuned to detect anomalies as well as adversarial attacks, preferably using some form of neural network. However, we propose a system that is initially and primarily designed and evaluated in this paper against user impersonation. To accomplish this, we consider two ideas: the first idea being that every user has a unique sEMG signature. *Aeles et al.* [12] mention finding consistent and unique signatures in how they recruit muscles for everyday actions like walking and pedaling, and that muscle activation timing and underlying neural drives are distinct to each person. *He et al.* [13] also confirm this idea of a unique EMG signature, mentioning that sEMG signals, when used in a biometric authentication context, could identify users with high accuracy, underlining the nature of EMG signals to be biologically unique in some way to an individual. The second idea is that what makes these signatures unique can be extracted by some form of machine learning to extract features that might include temporal, spatial, or frequency-based patterns, and can be used to identify and distinguish EMG users from one another. The previous section names a number of works that use machine learning models for sEMG-based authentication, so we know that training a model to recognize these unique signatures is feasible.

Additionally, we employ RMS maps as a core part of this model, taking direct inspiration from *Li et al.* [11] to turn this model from one designed around raw-sEMG signal analysis to a visual analysis task. Converting EMG signals to RMS maps allows us to compress our data, and in doing so, compress the dimensions of layers we would need in a neural network

to analyze this visual data, as well as significantly reduce the amount of computation that goes into training on this data.

While our work follows the use of [11]'s RMS maps closely, it differs in a large aspect: instead of using a deep learning framework, we utilize a convolutional autoencoder to determine the difference between a authorized user and an intruder by reconstructing an image.

To assemble this autoencoder, we create an encoder designed with convolutional layers to capture features from the RMS maps that embeds data to a latent space of only eight dimensions. From this latent space, a decoder performs transformations inverse to the encoder to reconstruct an RMS map.

Below is a table of our convolutional autoencoder structure, which holds a notably lightweight size of 10.22 KB (implemented via Tensorflow's Keras library). We designed this model around the Ninapro DB5 dataset, which contains 16 channels of sEMG from subjects using two Thalmic Myo armbands. These 16 channels are converted via a sliding window technique and RMS conversions to create a 2 by 8 two-dimensional heatmap of single color values from 0 to 1. The results of our autoencoder with the DB5 data is discussed later in our results section.

Layer	**Transformation**	**Output Dimensions**
1 (Input)	None	(2, 8, 1)
2 (Encoder)	2D Conv - 16 Filters - (2,2) Kernel - (1,1) Stride	(2, 8, 16)
3 (Encoder)	Batch Normalization	(2, 8, 16)
4 (Encoder)	2D Conv - 8 Filters - (2,2) Kernel - (1,2) Stride	(2, 4, 8)
5 (Encoder)	Flattening	(64)
6 (Latent)	Fully Connected (Dense) ReLU	(8)
7 (Decoder)	Fully Connected (Dense) ReLU	(64)
8 (Decoder)	Reshaping	(2, 4, 8)
9 (Decoder)	2D Conv Transpose - 8 Filters - (2,2) Kernel - (1,2) Stride	(2, 8, 16)
10 (Decoder)	2D Conv Transpose - 16 Filters - (2,2) Kernel - (1,1) Stride	(2, 8, 1)
11 (Output)	2D Conv - 1 Filter - (2,2) Kernel Size - (1,1) Stride	(2,8,1)

TABLE I
CONVOLUTIONAL AUTOENCODER LAYERS

We train the convolutional autoencoder on RMS-converted sEMG data of an authorized user to teach it what the movements and spatial intensity of their muscles look like, utilizing the mean squared error (MSE) between reconstructed images and the input image to serve as a trainable loss. This aims for the autoencoder to reconstruct images with a low MSE when RMS images similar to those of the authorized user are inputted, and also for a higher MSE when RMS images belong to an intruder, which to the model look dissimilar to the authorized user's RMS images, and results in a more vastly different reconstructed RMS image than what was inputted.

To create this autoencoder model, we used Keras from the Tensorflow library in Python, and employed all of our experiments via the Google Colaboratory platform on their given CPU and T4 runtime types. For data to test this model on, we used the Ninapro DB5 dataset, which contains 10 subjects performing 6 repetitions of 52 different movements, (which are three separate files separated into 1) basic finger movement, 2) isometric, isotonic hand configurations and basic wrist movements, and 3) gasping and functional movements). while wearing two Thalmic Myo armbands, each armband providing 8 channels of sEMG data sampled at 200 HZ. We converted this data to RMS maps by segmenting the data using a sliding window technique, where a 200 millisecond (or 40 sample) window across the 16 channels converts all of the instantaneous amplitudes in that range into a RMS values, creating 1 of 16 heatmap blocks per respective channel. After each channel's window has created its own heatmap block, we move this sliding window by 100 milliseconds (20 samples) forward in the data to get the next set of heatmap blocks for the next RMS map. We keep moving the sliding window until we reach the end of movement data file for a given subject.

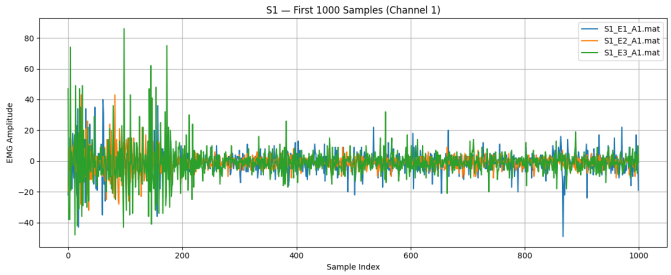

Fig. 1. Raw sEMG data of the first 1000 sample of Subject 1's three movement files.

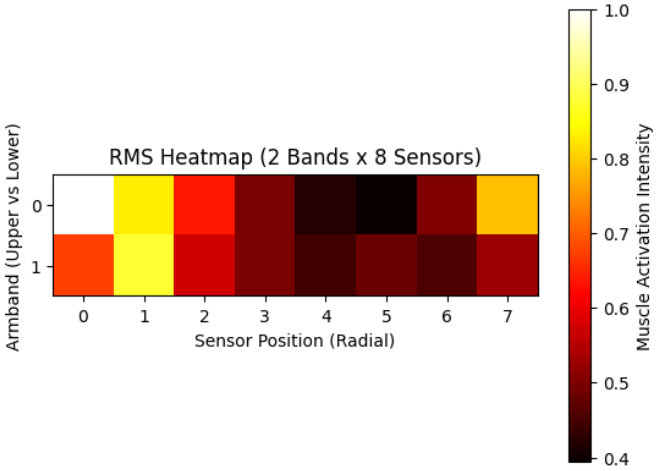

Fig. 2. A sample RMS map from Subject 1's First Movement File

To evaluate our model against that of an intruder's sEMG signals, we designed our first experiment based on training our model in the first two of the movement files of Subject 1 and then having the model reconstruct RMS heat maps from each of the RMS-converted movement files of Subject 2 and 3, as well as the last movement file of Subject 1 as a holdout test. To find the 'detection rate' that each set of heatmaps

had, we take each inputted RMS map in a set, and take its MSE against its respective reconstruction from the model. We push this set of MSEs through a percentile function that takes the 95% percentile of errors (where the 0th percentile is the smallest error and the 100th percentile is the largest error) and sets that specific percentile as the security threshold. Then, to get the detection rate for a set of movements, we simply calculate the average of the number of MSEs that are greater than the security threshold to get the percentage of RMS maps that are 'caught' by the model.

In our experiments, we validate the performance of our model's ability to differentiate between users by withholding different movement files (where the autoencoder is not trained on them) belonging to our authorized user, Subject 1. We test how well the autoencoder is able to recognize the authorized user from their untrained movements, as well as how well intruders with all of their movement files are evaluated against the model when it has trained on either all or some of the authorized user's movements.

IV. Results

We demonstrated that the proposed RMS-based intrusion detection system (IDS) can identify unique user sEMG signatures using two simultaneous data streams captured from Myo armbands placed at different positions along a single arm. Using RMS heatmap representations and an extremely lightweight 2D CNN autoencoder, we produced a model under 11 KB in size, which represents, to our knowledge, a novel and highly efficient approach to sEMG-based biometric identity detection. Particularly notable is the model's ability to run inferences on small data samples of 40 points, moving the system closer to near real-time detection compared with prior RMS-based methods that required 400 samples or roughly 2 full seconds (*Cedeño et al.*) [14].

The results of our experiments show that the proposed CNN autoencoder was relatively effective at distinguishing the authorized user (S1) from the intruders (S2 and S3), despite the model's compact architecture and the limited computational resources assumed for its deployment. As illustrated in Fig. 3, our detection rates revealed a clear separation between user classes with the intruder S3 being detected at a rate of 61.1% while S2 proving to be more challenging at 32.7%. The authorized user's holdout data set, consisting of gestures not seen during training, yielded a false detection rate of 20.4%. Although consecutive training iterations produced some variability caused by weight generation when training, the overall trend remained consistent with the autoencoder reconstructed S1's RMS maps with systematically lower error rates than those of intruders.

These findings were further supported by the distribution of reconstruction errors. Boxplots in Fig. 5 demonstrated that the authorized user exhibited a tightly clustered, low-error distribution, whereas both intruders produced significantly higher and more variable reconstruction errors, this pattern suggests that, even with limited training exposure, the autoencoder successfully learned features and characteristics unique to S1's sEMG activity.

However, the study also revealed several constraints and limitations that impacted performance. A primary limitation we faced was the scarcity and narrow scope of our data. Due to the lack of equipment to produce our own sEMG recordings, we relied on the best openly available dataset that met our requirements, which further restricted the diversity and quantity of usable samples. Our training set ultimately consisted of only two of the three available exercise gesture sets for S1, with the third set withheld exclusively for testing. This limited gesture variety constrained the model's ability to generalize across natural variabilities in sEMG behavior that would typically occur in real-world scenarios. Consequently, the authorized user's holdout samples exhibited higher reconstruction error than would be acceptable for a production-level continuous intruder detection system.

The dataset also lacked diversity in gesture types, number of subjects and repeated trials. sEMG signals in real-world settings vary significantly with posture, arm orientation, muscle fatigue, sensor displacement, and day-to-day behavioral factors. Because the dataset did not include redundant repetitions across a period of time or conditions, the model learned gesture-specific representations rather than a fuller spectrum of the authorized user's biometric signature. This also means that the autoencoder may learn patterns specific to the experimental task rather than learning a richer representation of individual biometric and behavioral patterns, consistent with the limitations noted in prior works.

Due to the segmented and gesture-specific nature of the Ninapro DB5 recordings that were limited by file, we also observed that our validation results varied noticeably depending on which of S1's gesture files were withheld for testing. This variability limited our ability to make stronger claims about S1 being consistently better recognized than the intruder subjects as displayed in Fig. 4 and highlights an inherent constraint of working with movement-partitioned datasets rather than true repeated measure biometric recordings.

Furthermore, we recognize that similar to [14], the reliance on RMS computation restricts the system from achieving true continuous real-time detection, as the model still requires a sample window of 40 data points to be collected before constructing an RMS image for inference. This requirement inherently introduces latency into our system because no prediction can occur until all samples in a window are collected. Shrinking the sample window size reduces this latency by allowing RMS maps to be generated more quickly, but it also limits the number of features available for extraction, which may in turn reduce model accuracy.

Despite these challenges, the results demonstrate that even with highly constrained data and computational resources, our proposed IDS can extract and utilize personalized sEMG features with promising accuracy. These initial outcomes support the feasibility of developing compact, real-time sEMG-based detection systems using RMS heatmaps and lightweight CNN models.

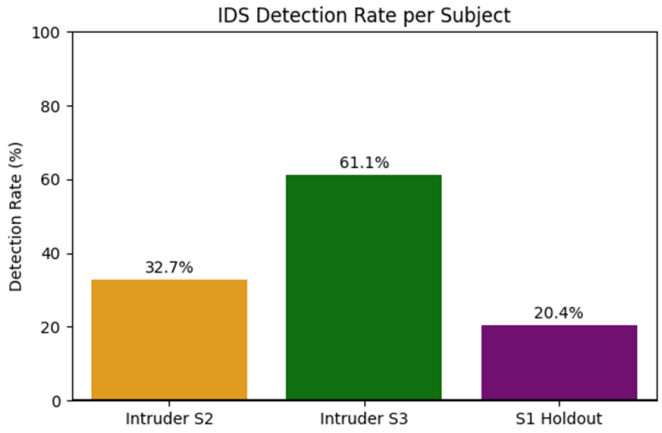

Fig. 3. Detection Rates of Intruders vs Authorized User's 3rd Movement File as Holdout.

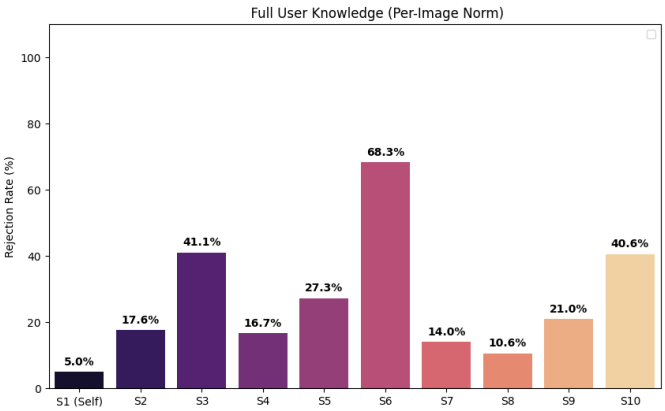

Fig. 4. Detection Rates of All Intruder Subjects vs Authorized User without Holdout.

Fig. 5. Boxplot of Reconstruction Errors.

V. Conclusion

In this study, we introduced an extremely lightweight 2D CNN autoencoder-based IDS capable of learning individualized sEMG signatures from RMS heatmaps while maintaining a model size of under 11 KB. The system demonstrated promising performance in distinguishing an authorized user from intruding subjects, even under constraints such as limited training data, reduced gesture diversity, and minimal computational complexity. The model's ability to operate on very small input samples also highlights its potential for near real-time biometric monitoring for wearable sEMG devices.

However, we faced several limitations that influenced performance and provided clear directions for improvement. The dataset's restricted size, lack of subject diversity, and absence of repeated gesture trials hindered our model's ability to learn and generalize across the full range of natural sEMG variability. Increasing the sample window sizes, incorporating denser feature extraction layers, and expanding the dataset to include more gestures, subjects, and repeated measurements would likely produce higher accuracy, at the cost of increased latency during inferencing and greater resource requirements. Additionally, RMS window requirements prevent true continuous detection, but exploring reduced window lengths represent a promising step toward real-time operation.

In the future, we hope to continue this work with a more comprehensive experimental framework that addresses both data quality and model robustness. A key priority for us is having the resources to create or somehow acquire richer sEMG datasets that include more subjects, a broader coverage of varying gestures, and repeated recordings collected across a longer period of time. Such data would enable the model to learn a more stable and realistic representation of sEMG behavior. Beyond improving data quality, future research should explore larger alternative autoencoder architectures, hybrid CNN transformer models, and optimized RMS window configurations that are able to balance latency with representational fidelity. Additional directions of research could include evaluating the system under known adversarial conditions such as radio frequencies or other signal injection attacks and examining its role within multi-user authorization settings or hierarchical IDS pipelines. Collectively, these extensions of our work would meaningfully advance the model toward real-world deployment and establish a stronger basis for robust continuous detection.

In the face of these limitations, this work still demonstrates the feasibility of using ultra-lightweight convolutional autoencoders applied to RMS-based sEMG representations for personalized anomaly detection. The methodology we present here lays the foundation for developing practical sEMG-based security systems that operate continuously on wearable devices.

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